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ODD END SEMESTER EXAMINATION NOV-DEC 2025

Name of The Program: **B.Tech (Petroleum Engineering)**

Semester: **V**

Name of The Course: **Petroleum Production Engg.-I**

Course Code: **TPE-503**

Time: 3 Hours

Maximum Marks: 100

Note:

- (i) All questions are Compulsory
- (ii) Answer any two sub questions among a, b & c in each Main Question.
- (iii) Total marks in each main question are twenty
- (iv) Each question Carries 10 marks.

Q1. (20 Marks) (CO-1,2)

- a) What is composition of petroleum and elemental composition of crude oil by weight percentage? Write classification of crude oil.
- b) Illustrate with the help of a diagram emulsion and types of emulsion. How the emulsions are formed and explain surface active agents.
- c) Write a short note on how produced water is treated in Petroleum Industry.

Q2. (20 Marks) (CO-2)

- a) Illustrate well head and why they are used in petroleum industry. Write its functions.
- b) With the help of Amine treater or Sweater explain the process of Gas sweetening and Also explain Glycol dehydration with the help of TEG.
- c) Write a short note on gas blanketing system and early production facility or system.

Q3. (20 Marks) (CO-2,3)

- a) Explain tank breathing system and tank battery. What do you understand by gauge hatches? Explain.
- b) Explain fixed roof tank and floating roof tank and Justify why Floating roof tank are best for petroleum industry. Differentiate between internal floating roof tank and internal floating roof tank.
- c) Describe what are the different methods by which we can control liquid leaks from tank and how leaks will be identified.

Q4. (20 Marks) (CO-4)

- a) Write the advantages and limitation of reciprocating separable compressors and reciprocating integral compressors
- b) Differentiate between positive displacement, dynamic and reciprocating compressors.
- c) What is a gas meter? Describe different types of gas meter.

Q5. (20 Marks) (CO-3,5)

- a) Explain what is water injection and describe the sources of injected water?
- b) Differentiate clearly between a two-phase separator and a three-phase separator and write its uses in petroleum industry.
- c) Explain demulsification? What are the different emulsion treating methods?

